z0p: A Zero-Dependency Transparent STARK Stack in Safe Rust

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Abstract

We present **z0p**, a complete zero-knowledge proving stack implemented from first principles in safe Rust with no dependencies beyond the compiler and the operating system. Binary fields, the BabyBear prime field, BLAKE3, Merkle trees, a parallel NTT, an arity-4 FRI polynomial commitment, four STARK systems (Fibonacci AIR, range checks, table membership, and hash-folded IVC), LogUp lookups, a minimal register-machine execution AIR, zero-knowledge masking, and a compact multilinear PCS prototype are all owned by a single crate under `#![forbid(unsafe_code)]`. The system is transparent (no trusted setup) and hash-based, hence plausibly post-quantum. Production parameters target a ≈ 100-bit query/proof-of-work schedule (rate 1/16, 40 queries, 20-bit grinding). FRI folds, STARK composition batches, `upcs`/Gemini points and betas, sumcheck rounds, and lookup identities draw challenges from a quartic extension $\mathbb{F}_{p^4}$, raising per-challenge algebraic bounds to ≈ 124 bits (see Section 6). Measured on a 16-core machine, a 1,048,576-step computation proves in 5.19 s ($\approx 4.95\,\mu$s/step), verifies in 2.3 ms, with a 206,353-byte proof. Validation combines 266 unit tests with three adversarial suites (≈ 800 hostile attempts per run, zero acceptances, zero panics), wire-exact size assertions on every benchmark, and machine-checked regression gates.

1 Introduction

Practical zero-knowledge infrastructure asks its users to trust a great deal: pairing-friendly elliptic curves, multi-party setup ceremonies, deep dependency trees, and provers whose soundness stories live in slide decks. At the same time, large-scale quantum computing threatens every discrete-logarithm and pairing-based proof system in deployment. The systems that survive that transition will be hash-based, transparently set up, small enough to audit, and tested hard enough to believe.

Contributions. z0p is, to our knowledge, the first *complete* STARK stack — fields, hash, Merkle trees, NTT, polynomial commitment, AIR systems, and zero-knowledge masking — implemented from scratch in safe Rust with zero dependencies:

- A transparent, plausibly post-quantum design: BLAKE3-256 digests and FRI Reed–Solomon commitments over BabyBear. No curves, no ceremonies.
- Production parameter sizing in one call (`Params::for_security(..., 100)`): blowup 16, queries, grinding, final layer, and cap height jointly optimized for a ≈ 100-bit query/PoW schedule. All algebraic challenges use a quartic extension field.
- Competitive scalar performance with parallelism over `std::thread` only: 1M-step proofs at ≈ 202k steps/s, millisecond verification, logarithmically growing proofs.
- Adversarial validation as a first-class artifact: every forgery class we could construct is tested continuously, including a fuzzer that once caught a genuine proof-malleability hole (since closed by binding dimensions into the transcript).

1.1 Related work

STARKs [2] and the FRI proximity test [3], strengthened by DEEP-FRI [4] and compiled via BCS-style Fiat–Shamir [1], are our direct lineage. Production STARK toolchains (notably Plonky2/Polygon Zero [8] and Risc Zero) share the BabyBear-plus-FRI engineering core but retain large dependency footprints. We differ in taking dependency-freedom and auditable minimalism as hard constraints, and in publishing adversarial test evidence alongside performance claims. Our lookups follow the LogUp logarithmic-derivative design [7]. The hash is BLAKE3 [6]; Fiat–Shamir is classical [5].

2 Preliminaries

Fields. Program words use binary fields $GF(2^8)$, $GF(2^{64})$, $GF(2^{128})$. Polynomial domains use BabyBear $\mathbb{F}_p$, $p = 2^{31} - 2^{27} + 1 = 2013265921$ with generator 31; $p - 1 = 2^{27} \cdot 15$, so 2^k-roots of unity exist for $k \leq 27$. Elements serialize as 4 canonical little-endian bytes; multiplication is Barrett reduction, inversion Fermat ($0 \mapsto 0$), batch inversion Montgomery's trick. FRI values and fold challenges live in the quartic tower $\mathbb{F}_{p^2} = \mathbb{F}_p[u]/(u^2 - 11)$, $\mathbb{F}_{p^4} = \mathbb{F}_{p^2}[v]/(v^2 - u)$; 11 is a quadratic non-residue and u a non-square; extension inversion uses the tower norm $(c + dv)^{-1} = (c - dv)/(c^2 - ud^2)$. Elements serialize as four canonical coordinates (16 bytes).

Domains. Trace domain $H = \{1, \omega, \ldots, \omega^{N-1}\}$; evaluation domain $D = \gamma \cdot \langle \Omega \rangle$ of size $M = N \cdot \beta$ (β the blowup), coset shift $\gamma = 31$; $M \leq 2^{27}$ enforced everywhere. Vanishing polynomial $Z_H(x) = x^N - 1$.

Hash and Merkle trees. Unkeyed BLAKE3-256, implemented from the specification and validated against reference vectors including chunk boundaries. Trees use $\text{leaf}(d) = H(0x00 \| d)$, $\text{node}(l, r) = H(0x01 \| l \| r)$ (second-preimage resistant). *Caps* expose an upper tree layer in the commitment so openings carry only middle path segments: column trees drop $\min(h, \text{depth})$ levels, FRI quad layers $\min(h, \text{depth} - 2)$ (the verifier rebuilds the bottom two levels from revealed values).

Transcripts. Append-only, length-prefix-framed Fiat–Shamir transcript with a full domain-separation schedule (FRI: `fri-dims` first, then per-digest `fri-layer-root`, three `fri-fold` challenges per layer, `fri-final`, `fri-pow`, and per-query `fri-query`; per-protocol `params`, `column`, and `alpha` labels; IVC `fold/step`, `fold/trace`, six `fold/alpha` per step plus per-step `fold/gamma`; lookup and sumcheck labels analogously). Proof-of-work grinding finds the smallest nonce with d leading zero bits under the history commitment; difficulties above 256 always fail.

3 FRI polynomial commitment

`commit` encodes n coefficients to $M = n\beta$ coset evaluations (shifted NTT); `commit_evals` takes caller-supplied LDE values of declared degree $\leq$ bound $< M$. Both enforce the rate gate bound $+ 1 \leq M/2$ at prove *and* verify time. Prover-echoed dimensions are absorbed under `fri-dims` before any challenge, closing proof malleability of those bytes.

Layers are bit-reversed; each adjacent quad sits at $(x, -x, j_0 x, -j_0 x)$ for the fixed fourth root j_0, sharing x^4. Writing $P(X) = A + BX + CX^2 + DX^3$ in $Y = X^4$, each layer folds as $A + r_0 B + r_1 C + r_2 D$ with fresh challenges $r_i \in \mathbb{F}_{p^4}$ (values embed base-field elements in the first coordinate), quartering the degree; point inverses are batch-inverted once per layer and folding is multi-threaded. Proofs carry the final layer in the clear plus, per query, all four values of every folded quad with middle path segments. The verifier replays the transcript, checks dimensions,

rate, cap shapes, and PoW, verifies the final layer by divided differences against residual bound bound $\gg 2f$, then checks every Merkle path and fold relation, chaining intermediate parents forward.

Conjecture 1 (FRI soundness sizing)**.** *Under the Johnson-radius heuristic ($\delta \approx 1 - \sqrt{\rho}$), rate $\rho = 1/8$ with 12 queries gives ≈ 18 bits (bench shape); rate $\rho = 1/16$ with 40 queries plus 20-bit grinding gives an $80 + 20 = 100$-bit query/PoW schedule (production shape). Caps affect size only. These counts exclude the finite-field algebraic term. FRI folds, the AIR/range/table/fold composition batches, `upcs`/Gemini evaluation points and fold betas, sumcheck rounds, and lookup's rational-identity challenge all draw $\mathbb{F}_{p^4}$ challenges, so a nonzero degree-d identity vanishes with probability about d/p^4 per challenge. The former approximately 31-bit algebraic ceiling is gone. The production schedule remains a query/PoW formula, not an end-to-end soundness proof.*

4 AIR systems

All application protocols share one pattern: interpolate columns, extend on the LDE coset, Merkle-commit joint rows, evaluate a composition vector (geometric vanishing recurrence with Montgomery-batched denominators, bit-identical to scalar evaluation), FRI-commit it, open trace columns at FRI query positions, and re-verify the identity per query.

Fibonacci AIR. Trace $A[0] = a_0$, $B[0] = b_0$, $A[i+1] = B[i]$, $B[i+1] = A[i] + B[i]$, claim $B[N-1] = b_{\text{last}}$. With $w_{\text{last}} = \omega^{N-1}$ and $Z_{\text{trans}}(x) = (x^N - 1)/(x - w_{\text{last}})$:

$$\begin{aligned} C(x) = \alpha_0 \frac{a' - b}{Z_{\text{trans}}(x)} + \alpha_1 \frac{b' - (a+b)}{Z_{\text{trans}}(x)} + \alpha_2 \frac{a - a_0}{x - 1} \\ + \alpha_3 \frac{b - b_0}{x - 1} + \alpha_4 \frac{b - b_{\text{last}}}{x - w_{\text{last}}}. \end{aligned} \tag{1}$$

Dividing (not multiplying) by Z_{trans} is load-bearing: the quotient is low-degree iff the transition holds on every covered row.

Range checks. Value column plus $n \le 24$ bit columns: decomposition $(V - \sum 2^j C_j)/Z_H$, per-bit boolean quotients $C_j(C_j - 1)/Z_H$, and a boundary quotient.

Table membership. $C(x) = \alpha_{\text{member}} \prod_{t \in T}(v - t)/Z_H(x) + \alpha_{\text{bound}}(v - \text{value})/(x - 1)$; small tables by design.

Folding (IVC). Chained segments share one batched FRI commitment over $\sum_s \gamma_s C_s$ with per-step challenges; one proof covers the chain.

Lookups. LogUp-style: multiplicities, β challenge, quotients $1/(\beta + w)$, $m/(\beta + t)$, rational-sum equality and four sumchecks; proving fails closed on out-of-table witnesses. A committed variant (`lookup_pcs`) commits the witness, table, multiplicities, and both quotient columns with BaseFold, binds every sumcheck final value with a point opening, and folds the total multiplicity weight into the sum-t claim with a fresh γ. At 1,024-in-1,024 with cap height 4 it proves in 367 ms, verifies in 21 ms, and carries 642,443 bytes. It is committed-input but not zero-knowledge, does not constrain columns to the base field, and pays for eight separate openings.

Program execution. The `vm` prototype proves execution of a public two-register straight-line program from a private initial state to a public output. Public one-hot selectors $\sigma_o(x)$ identify one of `Add`, `Mul`, `Swap`, `SetA`, `SetB`, or `Halt` per row; public immediate polynomials carry constants. With committed columns A, B, transition numerators T, U, and $Z_{\text{trans}}(x) = (x^n - 1)/(x - w_{\text{last}})$:

$$C(x) = \alpha_0 \frac{T(x)}{Z_{\text{trans}}(x)} + \alpha_1 \frac{U(x)}{Z_{\text{trans}}(x)} + \alpha_2 \frac{A(x) - a_{\text{last}}}{x - w_{\text{last}}} + \alpha_3 \frac{B(x) - b_{\text{last}}}{x - w_{\text{last}}}. \tag{2}$$

The final row has no outgoing transition. Composition degree is below $2n - 1$, so blowup is at least 8; production tuning uses 16. A 64-step $B16Q12C2$ plain proof proves in 1.5 ms, verifies in 0.24 ms, and carries 14,933 bytes. The plain path is not hiding; `prove_zk` masks both columns with random multiples of $x^n - 1$, preserving the trace and hiding query responses when $n_{\text{rand}} \geq Q$ (the caller sizes the final layer for the masked LDE). The program and output remain public and execution is straight-line; there is no control flow or memory.

Zero knowledge. Masked proving blinds columns with vanishing multiples from a ChaCha20 stream ($n_{\text{rand}} \geq$ queries enforced; key material zeroized on drop). Openings become witness-independent; verification is unchanged, and masked false statements still fail.

5 Encodings and parameters

Wire format (`VERSION` = 4): counts as $u32$, dimensions as $u64$, base fields as 4 canonical bytes and extension values as 16 canonical bytes ($< p$ per coordinate), digests as 32 bytes, direction bits 0/1. Lookup and sumcheck proofs use the same framing, with canonical extension values and nested versioned sumcheck bodies. The decoder fails closed on wrong versions, trailing bytes, truncation, non-canonical fields, and counts past hard caps (2^{20} queries, 64 folds, 128 path steps, 256 cap digests, 2^{27} elements, 2^{20} IVC steps) before any allocation. Validation: power-of-two domains, blowup ≥ 4 for AIR, M/final a power of four, $M \leq 2^{27}$.

6 Security analysis

Assumptions. BLAKE3 collision/preimage resistance; FRI soundness heuristic; Fiat–Shamir in the (Q)ROM. No discrete-log, pairing, or setup assumptions: Shor-inert by construction.

Enforced invariants. Bilateral rate gate; verifier-derived domains and query indices; final divided-difference check at true folded points; cap/path shape pins before hashing; `fri-dims` dimension binding; canonical-only decoding; PoW bound to history with queries sampled after.

Adversarial validation. Three layers: 266 unit tests (KATs, Horner cross-checks, parallel-vs-sequential identity); a 13-vector security suite (tampering of every query/fold/value/path/digest, 300 single-bit-flip fuzz trials, PoW forgery, rate gates both directions, statement replay); and a 10-vector red-team suite (cross-proof transplants, false statements, IVC surgery, codec-limit abuse, transcript abuse, exact rate boundaries, masked soundness, and a 500-input garbage storm that must fail closed without panicking). Standing: ≈ 800 hostile attempts per run, zero acceptances, zero panics, in debug and release.

7 Implementation and performance

Safe Rust, zero dependencies, parallelism over `std::thread` only (NTT stages $\geq 2^{16}$, folding, Merkle levels; all bit-identical). Key optimizations: Barrett field arithmetic, table-driven NTT, single-chunk hash fast paths, batched composition (geometric vanishing + Montgomery denominators), verifier on-demand domain points, Merkle caps. Measured 2026-10-20, release, medians, 16-core/31 GB, `rustc` 1.99.0:

Workload (production query/PoW schedule)	Prove	Verify	Size
AIR 1,024 steps	39 ms	1.2 ms	71,373 B
AIR 8,192 steps	81 ms	1.6 ms	104,033 B
AIR 65,536 steps	332 ms	2.1 ms	144,441 B
AIR **1,048,576** steps	**5.19 s**	**2.3 ms**	**206,353 B**
FRI poly-1024	71 ms	0.95 ms	43,174 B

Marginal cost $\approx 4.95\,\mu$s/step ($\approx$ 202k steps/s); proof sizes are deterministic and checked against the wire format every run. Verification is size-independent: each FRI domain point is derived on demand in $O(\log)$ field operations rather than from a materialized $O(\text{lde})$ layer table, so verify stays in the low milliseconds from 1,024 to 1,048,576 steps. Primitives: BLAKE3 1 MiB in 1.42 ms, BabyBear $\approx$ 136 Mops/s, NTT 2^{14} in 0.33 ms, 64k-leaf Merkle tree in 3.96 ms.

Compact multilinear PCS prototype. On 1,024 coefficients with blowup 16, 40 queries, and 20-bit grinding, the BaseFold-style prototype proves in 44 ms, verifies in 1.1 ms, and produces a 127,658-byte proof; cap height 4 gives 45 ms / 0.9 ms and 80,018 bytes. The Gemini v1 construction on the same shape takes 693 ms / 54 ms and 888,672 bytes. The prototype commits one root, reduces $f(r)$ to $c(\rho)$ with sumcheck, and uses Möbius FRI folds. Optional Merkle caps store a digest frontier per layer and hash it back to the layer root; decoder fuzzing covers both plain and capped proofs. It backs the committed lookup prototype but is not wired into the AIR stack and has no end-to-end soundness proof; treat these numbers as an engineering result, not an audit.

8 Limitations and future work

Bench defaults are fast but weak and must not be mistaken for production soundness. The table AIR is small-table by design; the committed LogUp prototype removes clear witness/table inputs but still uses eight separate BaseFold openings, lacks hiding, and does not enforce base-field column values. Vectorized hashing and the one-query recursion fragment are complete, FRI-backed univariate point and multi-point openings are in place, and Gemini plus a compact BaseFold-style prototype exist (one root, sumcheck plus Möbius FRI folds, optional Merkle caps). The prototype, committed lookup, and program-execution AIR are experimental and not wired into production AIR/STARK paths; batched openings, a hiding PCS/sumcheck, self-proving recursion, control flow, memory, instruction privacy, and a general zkVM remain roadmap items. The `vm` prototype is a public-program, private-initial-state, two-register straight-line machine with plain and masked proving; it has no memory or control flow. No external audit has yet been performed, for which the suites, gates, versioned wire format, and pinned baselines are the evidence package.

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